INCH-POUND

MIL-F-15733/49G
2 November 1988
SUPERSEDING
MIL-F-15733/49F
13 February 1985

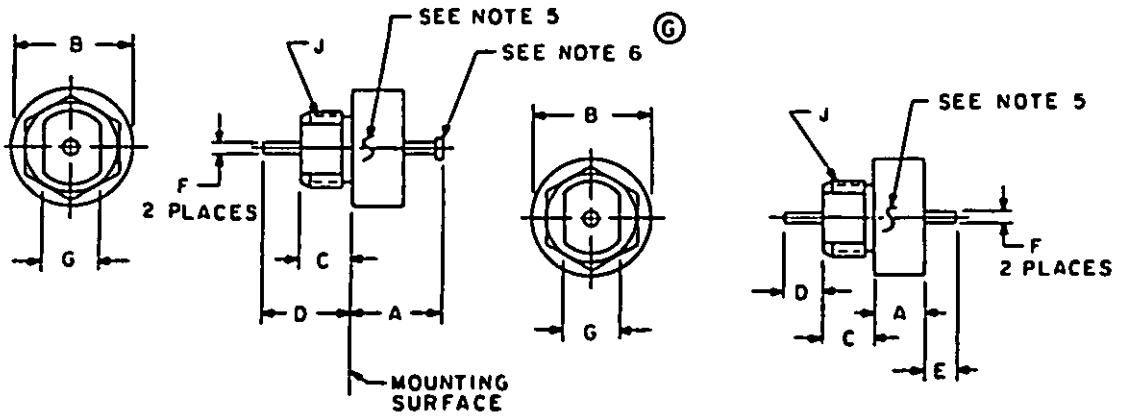
MILITARY SPECIFICATION SHEET

FILTERS AND CAPACITORS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL93

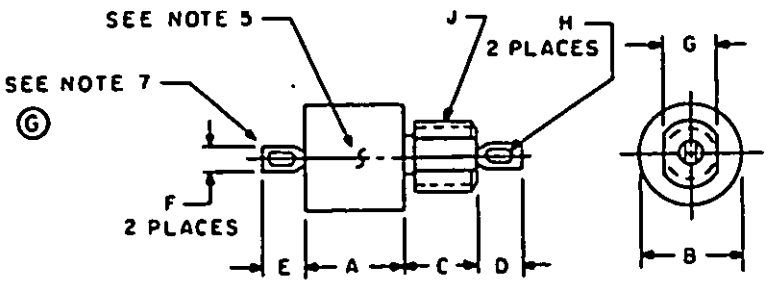
On 6 October 1980, P/N's M15733/49-0002, 0003, 0004, and 0005 were inactivated for new design. See table II, page 8, for replacement/supersession data.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-15733.



DASH NO. 0001 DASH NO. 0007



DASH NUMBERS 0002 THROUGH 0006, 0008 AND 0010

FIGURE 1. Case dimensions and circuit diagrams.

G denotes changes

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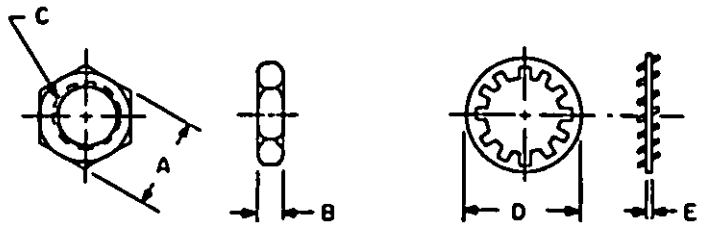
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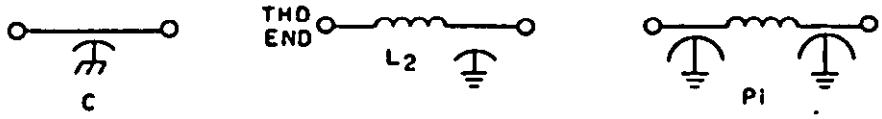
Dash no.	A		B		C		D		E		F	G		H	J
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Typ	Min	Max	Slot or hole ±.010	Mounting thread
Ⓖ 0001	.257 (6.53)	.319 (8.10)	---	.405 (10.29)	.177 (4.50)	.197 (5.00)	.281 (7.14)	.343 (8.71)	---	---	.040 (1.02)	.190 (4.83)	.210 (5.33)	---	.250-28 UNF-2A
0002 2/	.560 (14.22)	.580 (14.73)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	---	.190 (4.83)	.140 (3.56)	.160 (4.06)	.105 (2.67)	.190 (4.83)	.210 (5.33)	Slot .050 (1.27)	"
0003 2/	.160 (4.06)	.180 (4.57)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.160 (4.06)	---	.190 (4.83)	.105 (2.67)	.190 (4.83)	.210 (5.33)	.060 dia 1/ (1.52) both ends	"
0004 2/	.160 (4.06)	.180 (4.57)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.160 (4.06)	---	.190 (4.83)	.105 (2.67)	.190 (4.83)	.210 (5.33)	.060 dia 1/ (1.52) both ends	"
0005 2/	.210 (5.33)	.230 (5.84)	---	.385 (9.78)	.302 (7.67)	.322 (8.18)	.140 (3.56)	.160 (4.06)	---	.190 (4.83)	.105 (2.67)	.190 (4.83)	.210 (5.33)	.050 x .070 (1.27) x (1.78) or .070 (1.78) dia	"
Ⓖ 0006	.160 (4.06)	.180 (4.57)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.160 (4.06)	.100 (2.54)	.190 (4.83)	.105 (2.67)	.190 (4.83)	.210 (5.33)	.050 x .070 (1.27) x (1.78) or .070 (1.78) dia	"
0007	.160 (4.06)	.180 (4.57)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.250 (6.35)	.270 (6.86)	.250 (6.35)	.270 (6.86)	.040 (1.02)	.190 (4.83)	.210 (5.33)	---	"
Ⓖ 0008	.406 (10.31)	.416 (10.57)	---	.400 (10.16)	.180 (4.57)	.200 (5.08)	.140 (3.56)	.160 (4.06)	.140 (3.56)	.160 (4.06)	---	.190 (4.83)	.210 (5.33)	.050 x .070 (1.27) x (1.78) or .070 (1.78) dia	"
Ⓖ 0010	.500 (12.70)	.520 (13.21)	---	.400 (10.16)	.180 (4.57)	.200 (5.08)	.140 (3.56)	.160 (4.06)	.140 (3.56)	.160 (4.06)	---	.190 (4.83)	.210 (5.33)	.050 x .070 (1.27) x (1.78) or .070 (1.78) dia	"

1/ Slot optional.
2/ Inactive for new design.

FIGURE 1. Case dimensions and circuit diagrams - Continued.

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CIRCUIT DIAGRAMS

Filter dash no.	Hex nut					Lockwasher			
	A		B		C	D		E	
	Min	Max	Min	Max	Thread	Min	Max	Min	Max
0001	.302 (7.67)	.322 (8.18)	.085 (2.16)	.095 (2.41)	.250-28 UNF-2B	.396 (10.06)	.408 (10.36)	.023 (0.58)	.027 (0.69)
0002	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	"	"	"	.018 (0.46)	.022 (0.56)
0003	"	"	"	"	"	"	"	"	"
0004	"	"	"	"	"	"	"	"	"
0005	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	"	"	"	"	"
0006	.302 (7.67)	.322 (8.18)	.085 (2.16)	.095 (2.41)	"	"	"	"	"
0007	.302 (7.67)	.322 (8.18)	.085 (2.16)	.095 (2.41)	"	"	"	"	"
0008	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	"	"	"	"	"
0010	- - -	"	- - -	"	"	"	"	"	"

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagrams are for information only.
4. All filters shall be supplied with mounting hardware.
5. Terminal identification. The case shall be marked at the threaded end of the filter, with the symbol "L", for dash numbers 0001, 0003, 0004, 0006 and 0007.
6. Turret head is optional.
7. Flag terminals shall be aligned within 10° of the vertical (threaded bushing flats).

FIGURE 1. Case dimensions and circuit diagrams - Continued.

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TABLE I. Electrical characteristics.

Dash no.	Circuit diagram	Max rated current (amperes)	Rated voltage (volts)		Max voltage drop (volts, dc)	Minimum insertion loss (dB) in accordance with MIL-STD-220 at +25°C 1/									Cap. -0 +100% (μF)
			DC	AC 6/		30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	5 GHz	10 GHz	
(G) 0001	L_2	15	50 4/	---	.15	8	20	28	38	55	70	70	70	70	0.68
0002 2/	P1	3	150 7/	---	.15	---	7	24	54	80	80	80	---	80	---
0003 2/	L_2	10	50 7/	---	.08	15	28	33	44	60	---	70	---	---	---
0004 2/	L_2	10	50 3/	---	.08	15	28	33	44	60	---	70	---	---	---
0005 2/	P1	15	50 5/	---	.09	70 dB from 420 MHz to 1.0 GHz, inclusive									---
0006	L_2	15	50 4/	---	.15	15	28	33	44	60	---	70	---	70	1.2
0007	L_2	15	50 4/	---	.15	15	28	33	44	60	---	70	---	70	1.2
(G) 0008	C	15	50 7/	---	.06	---	33	40	50	65	---	70	---	---	2.1
(G) 0010	C	15	330 7/	125	.06	---	2	7	17	37	---	70	---	---	0.062

1/ Insertion loss measurements shall be made under full load over the frequency range of 150 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no load.

2/ Inactive for new design.

3/ Voltage rating at +125°C (Note - derated from 80 V dc at +85°C) for dash number 0004.

4/ Voltage rating at +125°C (Note - derated from 100 V dc at +85°C) for dash numbers 0001, 0006, and 0007.

5/ Voltage rating at +85°C.

6/ AC voltage rating shall be dc to 400 Hz.

(G) 7/ Voltage rating at +125°C.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Weight: 8.5 grams, maximum.

(6) Case: Metal, silver-plated finish (-0001); metal tin-plated in accordance with MIL-T-10727 or electro-tin fused or, tin-lead plated in accordance with MIL-P-81728, or hot-solder dipped (40 percent to 60 percent tin in accordance with QQ-S-571) (-0002 through -0008, and -0010).

Seal: Glass-to-metal.

Terminals: Solderable (see figure 1).

Dielectric: Ceramic.

(6) Operating temperature range: -55°C to +125°C, temperature characteristic D (-0001 through -0004, -0006 through -0008, and -0010) and -55°C to +85°C, temperature characteristic E (-0005).

Rated voltage: See table I.

Rated current: See table I.

Insertion loss (at 25°C): Shall be as specified in table I.
(at -55°C and +125°C): Insertion loss shall be as specified in table I except as follows:

Up to 10 MHz: A degradation of 4 dB from the value specified in table I shall be allowed.

Creepage and clearance: Not applicable.

Voltage conditioning (applicable to -0001, quality conformance inspection only): Prior to group A inspection, 100 percent of each lot of filters shall be subjected to voltage conditioning as follows:

Test temperature: +125°C +4°C, -0°C.

Test voltage: 140 percent of the rated dc voltage.

Points of application of test voltage: Between the case and either terminal.

Duration of exposure to test voltage: 250 hours, minimum.

Measurements at +25°C after the test:

Dielectric withstanding voltage: 140 percent of rated dc voltage shall be applied for 1 minute.

Insulation resistance: Shall be not less than 500 megohms.

Insertion loss (check test): Shall meet the initial requirements.

A reject shall be defined as a filter that fails the dielectric withstanding voltage test, insulation resistance requirements, or insertion loss (check test). If the total rejects from any particular lot exceed 10 percent, the entire lot shall be rejected.

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Radiographic inspection (applicable to -0001, quality conformance inspection only): Prior to group A inspection, and following voltage conditioning, 100 percent of each lot of filters shall be subjected to radiographic inspection as follows:

Filters shall be X-rayed in one plane at 90-degree rotation (perpendicular to the filter longitudinal axis). Any evidence of poor capacitor mounting or uneven soldering of capacitor to case shall be cause for rejection. Rejected units shall be dissected and examined for poor workmanship or uneven soldering.

Any defects noted shall require corrective action prior to acceptance of any future lots. Magnification during visual examination shall be two power minimum. Unless otherwise specified, one radiograph shall be taken of each unit. Filters shall be positioned to yield a longitudinal image of the body.

(G) Seal: Method 112, MIL-STD-202, test condition A (-0001 through -0004, -0006 through -0008, and -0010), and test condition A (-0005).

Capacitance to ground: See table I.

Capacitance tolerance: +100 percent, -0 percent.

Temperature rise: +25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.

Test temperature: +25°C.

(G) Test voltage: 125 percent of rated voltage (-0001); 100 V dc (-0003 and -0004); 370 V dc (-0002); 200 V dc (-0005); and 200 percent of rated voltage (-0006, -0007, -0008, and -0010) applied between terminals and case for 1 minute; polarity of the test voltage shall then be reversed and the voltage shall be applied for 1 minute.

Barometric pressure (reduced): Method 105, test condition B (50,000 feet), MIL-STD-202.

Dielectric withstanding voltage: 125 percent of the rated voltage applied for 1 minute.

(G) Insulation resistance: Method 302, MIL-STD-202.

Test temperature: +25°C.

Test potential: Rated dc voltage or 100 V dc whichever is less.

Insulation resistance shall be not less than:

Dash numbers	Megohms
0001	500
0002 through 0005, 0008	100
0006, 0007, and 0010	1,000

Dissipation factor (-0008 and -0010): 3 percent.

Voltage drop: See table I.

Insertion loss: In accordance with MIL-STD-220 and table I.

Insertion loss (check test): In accordance with MIL-F-15733.

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Overload: In accordance with MIL-F-15733.

Measurements at +25°C after test:

Insulation resistance: Shall meet initial requirements.

Terminal strength: Method 211, MIL-STD-202.

Test condition A (pull): Applied force, 5 pounds (-0001 through -0007); 4.5 pounds +3, -0 ounces (-0008 and -0010).

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

(G) Thermal shock and immersion:

Thermal shock: Method 107, test condition A, MIL-STD-202 (-0005); and method 107, test condition A, MIL-STD-202 (-0001 through -0004, -0006 through -0008, and -0010), except the temperature at step 3 shall be +125°C.

Immersion: Method 104, MIL-STD-202, test condition A (-0002 through -0008, and -0010), and test condition B (-0001).

Measurements after final cycling:

Dielectric withstanding voltage: 90 percent of the initial voltage shall be applied for 5 ±1 seconds.

Insulation resistance: Shall be not less than 30 percent of the initial requirements.

Insertion loss (check test): Shall meet the initial requirements.

Capacitance change: Not more than ±20 percent for dash numbers -0008 and -0010.

(G) Shock (specified pulse): Method 213, MIL-STD-202, test condition I (-0001, -0002, -0003, -0004, -0006 through -0008, and -0010); and test condition A (-0005).

Electrical load: 100 percent of the rated dc voltage shall be applied between terminals and case.

Measurement after test: Insertion loss (check test) shall meet the initial requirements.

(G) Vibration, high frequency: Method 204, MIL-STD-202, test condition D (20 g's).

Electrical load: 100 percent of the rated dc voltage shall be applied continuously during test.

Temperature conditioning (-0002 through -0005 only): Prior to vibration in each plane, filters shall be placed in an oven and heated to +125°C plus one-half the temperature rise (-0002 through -0004) and to +85°C plus one-half the temperature rise (-0005). Vibration in each plane shall begin within 5 minutes after removal of the filters from the oven.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent open or short-circuiting.

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After vibration test: Filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Ⓖ Moisture resistance: Method 107, test condition A, MIL-STD-202 (-0005) and method 107, test condition A, MIL-STD-202 (-0001 through -0004, -0006 through -0008, and -0010), except the temperature of step 3 shall be +125°C. No measurements shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.

Polarization voltage: 50 volts dc.

Measurements after 24-hour drying period:

Dielectric withstanding voltage: 90 percent of the initial voltage shall be applied for 5 ±1 seconds.

Insulation resistance: Shall meet the initial requirements.

Insertion loss (check test): Shall meet the initial requirements.

Capacitance change: Not more than ±20 percent from initial measurement for dash nos. -0008 and -0010.

Ⓖ Life: In accordance with MIL-F-15733.

Test condition:

For qualification inspection:

Ⓖ D (1,000 hours).

For group C inspection:

Ⓖ B (250 hours) for -0001, -0003 through -0008, and -0010; D (1000 hours) for -0002.

Test temperature: +125°C (-0001 through -0004, -0006 through -0008, and -0010), and +85°C (-0005).

Test voltage: 1.4 times rated dc voltage.

Measurements after test:

Dielectric withstanding voltage: 90 percent of the initial voltage shall be applied for 5 ±1 seconds.

Insulation resistance: Shall be not less than 30 percent of the initial requirements.

Insertion loss (check test): Shall meet the initial requirements.

Marking: Filters shall be marked in accordance with method I of MIL-STD-1285 and as follows:

Minimum marking (-0001): Filters shall be marked with the military part number and terminal identification, as a minimum. The full marking specified in MIL-F-15733 shall be marked on the unit package.

Full marking (-0002): Filters shall be marked with the full marking specified in MIL-F-15733.

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Minimum marking (-0003 through -0008, and -0010): Filters shall be marked with the military part number, source code, date code, and terminal identification. The full marking specified in MIL-F-15733 shall be marked on the unit package.

Part identification number: M15733/49-(dash number from table I).

Ⓖ Initial qualification to MIL-F-15733/49 may be granted based on qualification to MIL-F-28861/1 or MIL-C-83439/2 as indicated in table II. Extension of qualification from MIL-F-28861/1 or MIL-C-83439/2 is permissible under the following provisions:

a. The MIL-F-15733/49 parts use the same design and dielectric characteristics as the comparable MIL-F-28861/1 or MIL-C-83439/2 parts.

b. The MIL-F-28861/1 or MIL-C-83439/2 data (as applicable) verifies that the physical and electrical characteristics of the MIL-F-15733/49 parts are satisfied.

Ⓖ TABLE II. Extension of qualification.

Qualification to MIL-F-28861/1	Will qualify MIL-F-15733/49	Qualification to MIL-C-83439/2	Will qualify MIL-F-15733/49
-001	-0003	-001	-0008
-001	-0004		
-001	-0006	-003	-0010

Supersession data: See table III.

Ⓖ TABLE III. Supersession data.

Superseded military part identification number (old)	Superseding military part identification number (new)
M83439/02-001	M15733/49-0008
M83439/02-003	M15733/49-0010
M15733/49-0002	M15733/38-0003
M15733/49-0003	M15733/49-0006
M15733/49-0004	M15733/49-0006
M15733/49-0005	M15733/38-0007

Ⓖ Part identification number M15733/49-0009 has been deleted. For new design, use M28861/01-010 of MIL-F-28861/1.

Cataloging information: Dash numbers 0008 and 0010 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 0001 through 0007 shall be cataloged under FSC 5915 as radio frequency interference filters.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0063)

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